

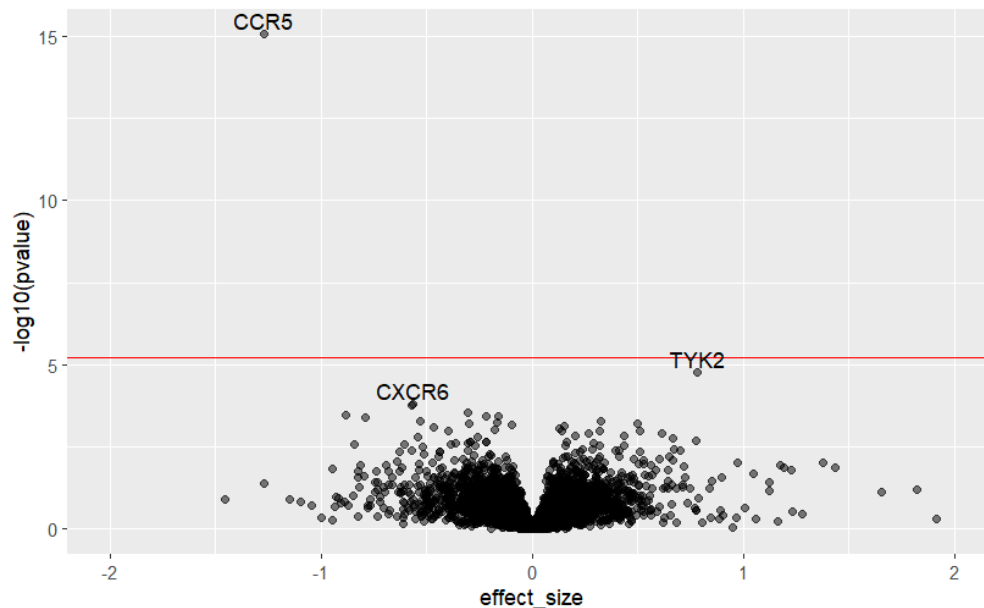
## Methods

In this assignment, we ran a transcriptome-wide analysis study on a particular GWAS performed by the COVID-19 Host Genetics Initiative, testing whether each gene's expression level is associated with severe COVID-19 respiratory symptoms. The analysis was performed by running S-PrediXcan using the summary statistics of the provided GWAS. This TWAS took the GWAS summary statistics, a tissue-specific prediction model, and its corresponding covariate LD file as inputs, and outputted a .csv file recording the p-value and effect size of each gene. Lung tissue was chosen for analysis, since COVID-19 is a respiratory disease that most severely affects breathing and pulmonary function.

After collecting TWAS data, an overrepresentation analysis was performed on only genes with  $p < 0.01$  using FUMA's GENE2FUNC. We hypothesize that more genes in the TWAS falls into a certain gene set than expected, which would indicate enrichment or overrepresentation in a specific biological pathway or cell type.

## Results

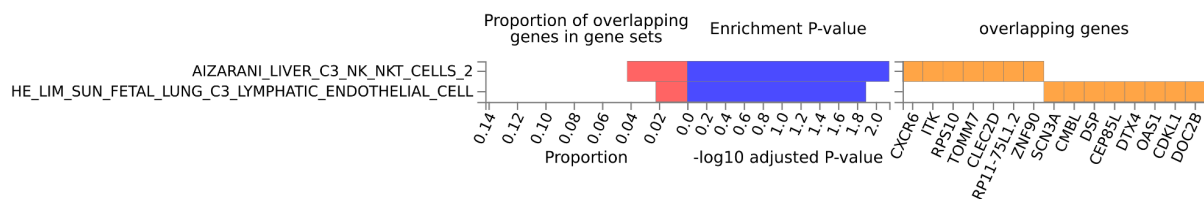
Overall, 83% of the prediction model's SNPs were used in this TWAS. 7930 genes were tested in total, with its Bonferroni-adjusted significance threshold at  $6.30517e-06$ . After creating a volcano plot, we identified the top three genes as CCR5, TYK2, CXCR6.



CCR5 is the top most significant gene with a p-value of 8.499364e-16 and effect size of -1.2718632. CCR5 is a [chemokine gene involved in inflammatory processes](#), and the negative effect size from our results suggests that the presence of CCR5 reduces the risk of severe COVID-19. A paper from [Files et al.](#) observes that inhibiting CCR5 pathways could reduce the inflammation that occurs in COVID-19 patients, which is logical with our results— CCR5 is an immunoregulatory gene which would naturally be protective against COVID-19, but it can cause heavy inflammation, fibrosis, and cytokine storms in severe cases of COVID-19 and thus, as Files et al. and other studies suggests, would need to be inhibited to prevent worsening of the disease.

TYK2 is the second most significant gene with a p-value of 1.734110e-05 and effect size of 0.7828881. TYK2 [encodes a member of the tyrosine kinase and may contribute to anti-viral immunity](#), though its positive effect size in our data suggests that its presence is a mild risk factor for COVID-19. This is consistent with the findings of [Rizi et al.](#), who establish that particular SNPs of TYK2 may be a risk factor for disease severity of COVID-19.

CXCR6 is the third most significant gene with a p-value of 1.623082e-04 and effect size of -0.5670695. CXCR6 [maps to the chemokine receptor gene cluster and is expressed in certain T lymphocyte subsets](#). [Dai et al.](#) finds that the “CXCR6 and CCR9... have a protective effect on the lung and a risk effect on whole blood, respectively” which is consistent with the negative effect size that we observe in our data.



Lastly, this is the FUMA plot created from a list of 102 significant ( $p < 0.01$ ) genes inputted into GENE2FUNC, 101 of which FUMA recognized as having valid Ensembl IDs. One gene set category was significant after adjustment— *Cell type signature* (*MsigDB c8*), containing two significant gene sets, *AIZARANI\_LIVER\_C3\_NK\_NKT\_CELLS\_2* (p-value = 8.70e-6, containing 142 genes) and *HE\_LIM\_SUN\_FETAL\_LUNG\_C3\_LYMPHATIC\_ENDOTHELIAL\_CELL* (p-value = 3.06e-5, containing 362 genes). The NK/NKT cells had 7 genes overlapping with my gene set, and the fetal lung lymphatic endothelial cells had 8 genes overlapping. These results are consistent with the rest of our TWAS findings, showing enrichment in lung tissue cells and cells involved in immune response.